Audience: LEWG, SG14, WG21
Document number: D0447R13
Date: 2021-03-25
Project: Introduction of std::colony to the standard library
Reply-to: Matthew Bentley <mattreecebentley@gmail.com>

Introduction of std::colony to the standard library

Table of Contents

I. Introduction

II. Questions for the committee

III. Motivation and Scope

IV. Impact On the Standard

V. Design Decisions

VI. Technical Specification

VII. Acknowledgements

VIII. Appendixes:

A. Basic usage examples

B. Reference implementation benchmarks

C. Frequently Asked Questions

D. Specific responses to previous committee feedback

E. Typical game engine requirements

F. Time complexity requirement explanations

G. Why not constexpr?

Revision history

- R13: Revisions based on committee feedback. Skipfield template parameter changed to priority enum in order to not over-specify container implementation. Other wording changes to reduce over-specifying implementation. Some non-member template functions moved to be friend functions. std::limits changed to std::colony_limits. block_limits() changed to block_capacity_limits().
- R12: Fill, range and initializer_list inserts changed to void return, since the insertions are not guaranteed to be sequential in terms of colony order and therefore returning an iterator to the first insertion is not useful. Non-default-value fill constructor changed to non-explicit to match other std:: containers. Correction to reserve() wording. Other minor corrections and clarity improvements.
- R11: Overhaul of technical specification to be more 'wording-like'. Minor alterations & clarifications. Additional alternative approach added to Design Decisions under skipfield information. Overall rewording. Reordering based on feedback. Removal of some easily-replicated 'helper' functions. Change to noexcept guarantees. Assign added. get_block_capacity_limits and set_block_capacity_limits functions renamed to block_limits and reshape. Addition of block-limits default constructors. Reserve() and shrink_to_fit() reintroduced. Trim(), erase and erase_if overloads added.
- R10: Additional information about time complexity requirements added to appendix, some minor corrections to time complexity info. The 'bentley pattern' (this was always a temporary name) is renamed to the more astute 'low-complexity jump-counting pattern'. Likewise the 'advanced jump-counting skipfield' is renamed to the 'high-complexity jump-counting pattern' - for reasoning behind this go here. Both refer to time complexity of operations, as opposed to algorithmic complexity. Some other corrections.

- R9: Link to Bentley pattern paper added, and is spellchecked now.
- R8: Correction to SIMD info. Correction to structure (missing appendices title, member functions and technical specification were conjoined, acknowledgments section had mysteriously gone missing since an earlier version, now restored and updated). Update intro. HTML corrections.
- R7: Minor changes to member functions.
- R6: Re-write. Reserve() and shrink_to_fit() removed from specification.
- R5: Additional note for reserve, re-write of introduction.
- R4: Addition of revision history and review feedback appendices. General rewording. Cutting of some dead wood. Addition of some more dead wood. Reversion to HTML, benchmarks moved to external URL, based on feedback. Change of font to Times New Roman based on looking at what other papers were using, though I did briefly consider Comic Sans. Change to insert specifications.
- R3: Jonathan Wakely's extensive technical critique has been actioned on, in both documentation and the reference implementation. "Be clearer about what operations this supports, early in the paper." - done (V. Technical Specifications). "Be clear about the O() time of each operation, early in the paper." - done for main operations, see V. Technical Specifications. Responses to some other feedbacks included in the foreword.
- R2: Rewording.

I. Introduction

The purpose of a container in the standard library cannot be to provide the optimal solution for all scenarios. Inevitably in fields such as high-performance trading or gaming, the optimal solution within critical loops will be a custom-made one that fits that scenario perfectly. However, outside of the most critical of hot paths, there is a wide range of application for more generalized solutions.

Colony is a formalisation, extension and optimization of what is typically known as a 'bucket array' container in game programming circles; similar structures exist in various incarnations across the high-performance computing, high performance trading, 3D simulation, physics simulation, robotics, server/client application and particle simulation fields (see: https://groups.google.com/a/isocpp.org/forum/#!topic/sg14/1iWHyVnsLBQ).

The concept of a bucket array is: you have multiple memory blocks of elements, and a boolean token for each element which denotes whether or not that element is 'active' or 'erased', commonly known as a skipfield. If it is 'erased', it is skipped over during iteration. When all elements in a block are erased, the block is removed, so that iteration does not lose performance by having to skip empty blocks. If an insertion occurs when all the blocks are full, a new memory block is allocated.

The advantages of this structure are as follows: because a skipfield is used, no reallocation of elements is necessary upon erasure. Because the structure uses multiple memory blocks, insertions to a full container also do not trigger reallocations. This means that element memory locations stay stable and iterators stay valid regardless of erasure/insertion. This is highly desirable, for example, in game programming because there are usually multiple elements in different containers which need to reference each other during gameplay and elements are being inserted or erased in real time.

Problematic aspects of a typical bucket array are that they tend to have a fixed memory block size, do not re-use memory locations from erased elements, and utilize a boolean skipfield. The fixed block size (as opposed to block sizes with a growth factor) and lack of erased-element re-use leads to far more allocations/deallocations than is necessary. Given that allocation is a costly operation in most operating systems, this becomes important in performance-critical environments. The boolean skipfield makes iteration time complexity undefined, as there is no way of knowing ahead of time how many erased elements occur between any two non-erased elements. This can create variable latency during iteration. It also requires branching code, which may cause issues on processors with deep pipelines and poor branch-prediction failure performance.

A colony uses a non-boolean method for skipping erased elements, which allows for O(1) amortized iteration time complexity and more-predictable iteration performance than a bucket array. It also utilizes a growth factor for memory blocks and reuses erased element locations upon insertion, which leads to fewer allocations/reallocations. Because it reuses erased element memory space, the exact location of insertion is undefined, unless no erasures have occurred or an equal number of erasures and insertions have occurred (in which case the insertion location is the back of the container). The container is therefore considered unordered but sortable. Lastly, because there is no way of predicting in advance where erasures ('skips') may occur during iteration, an O(1) time complexity [ ] operator is not necessarily possible (depending on implementation) and therefore, the container is bidirectional but not random-access.

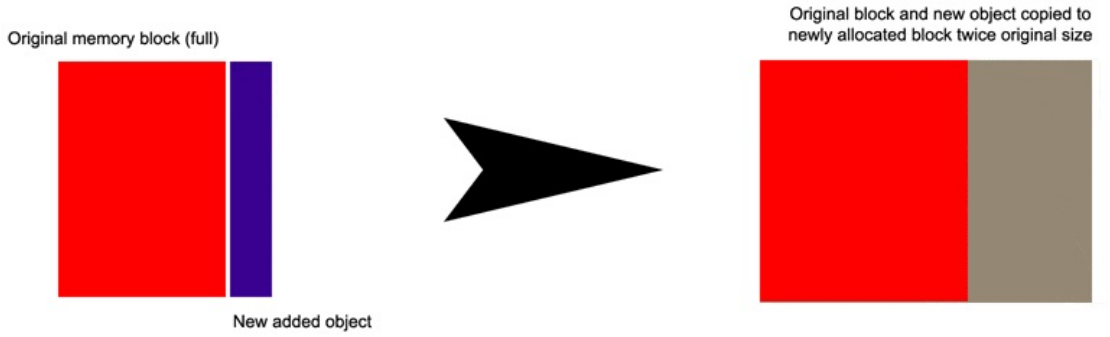
There are two patterns for accessing stored elements in a colony: the first is to iterate over the container and process each element (or skip some elements using the advance/prev/next/iterator ++/-- functions). The second is to store the

iterator returned by the insert() function (or a pointer derived from the iterator) in some other structure and access the inserted element in that way. To better understand how insertion and erasure work in a colony, see the following images.

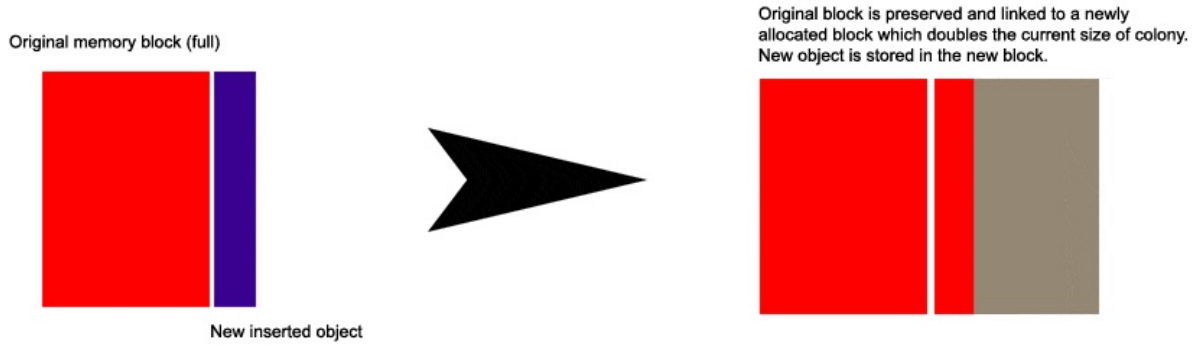
Insertion to back

The following images demonstrate how insertion works in a colony compared to a vector when size == capacity.

Vector:



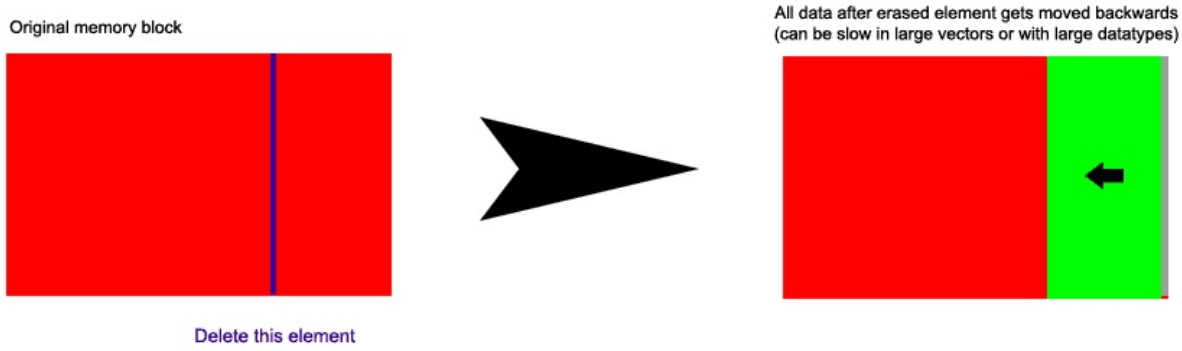
Colony:



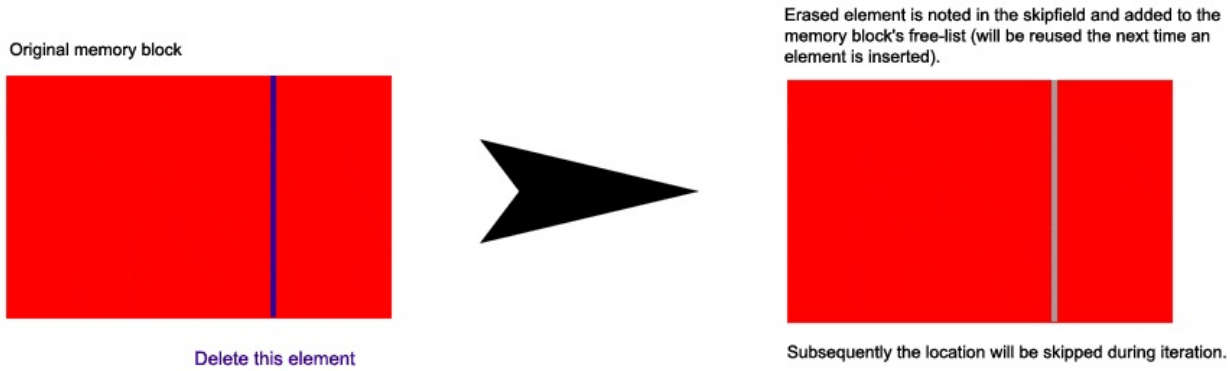
Non-back erasure

The following images demonstrate how non-back erasure works in a colony compared to a vector.

Vector:



Colony:



II. Questions for the Committee

1. It is possible to make the memory() function constant time at a cost (see details in it's entry in the design decisions section of the paper) but since this is expected to be a seldom-used function I've decided not to do so and leave the time complexity as implementation-defined. If there are any objections, please state them. Also, memory_usage() has been suggested as a better name?
2. The conditions under which memory blocks are retained by the erase() functions and added to the "reserved" pile instead of deallocated, is presently implementation-defined. Are there any objections to this? Should we define this? See the notes on erase() in Design Decisions, and the item in the FAQ. One option is to specify that only the current back block may be retained, however I feel like this should be implementation-defined.
3. Given that this is a largely unordered container, should resize() be included? Currently it is not and I see no particular reason to do so, but if there are valid reasons let me know.
4. Currently get_iterator_from_pointer() takes a pointer, should it be overloaded to take either a const_pointer or pointer and return a const_iterator or iterator respectively? This issue has been raised by several users due to the

difficulties of using colony with client side code. Also, happy to get a much more succinct name for that function if one exists.

III. Motivation and Scope

Note: Throughout this document I will use the term 'link' to denote any form of referencing between elements whether it be via ids/iterators/pointers/indexes/references/etc.

There are situations where data is heavily interlinked, iterated over frequently, and changing often. An example is the typical video game engine. Most games will have a central generic 'entity' or 'actor' class, regardless of their overall schema (an entity class does not imply an ECS). Entity/actor objects tend to be 'has a'-style objects rather than 'is a'-style objects, which link to, rather than contain, shared resources like sprites, sounds and so on. Those shared resources are usually located in separate containers/arrays so that they can re-used by multiple entities. Entities are in turn referenced by other structures within a game engine, such as quadtrees/octrees, level structures, and so on.

Entities may be erased at any time (for example, a wall gets destroyed and no longer is required to be processed by the game's engine, so is erased) and new entities inserted (for example, a new enemy is spawned). While this is all happening the links between entities, resources and superstructures such as levels and quadtrees, must stay valid in order for the game to run. The order of the entities and resources themselves within the containers is, in the context of a game, typically unimportant, so an unordered container is okay.

Unfortunately the container with the best iteration performance in the standard library, vector[1], loses pointer validity to elements within it upon insertion, and pointer/index validity upon erasure. This tends to lead to sophisticated and often restrictive workarounds when developers attempt to utilize vector or similar containers under the above circumstances.

std::list and the like are not suitable due to their poor locality, which leads to poor cache performance during iteration. This is however an ideal situation for a container such as colony, which has a high degree of locality. Even though that locality can be punctuated by gaps from erased elements, it still works out better in terms of iteration performance[1] than every existing standard library container other than deque/vector, regardless of the ratio of erased to non-erased elements.

Some more specific requirements for containers in the context of game development are listed in the appendix.

As another example, particle simulation (weather, physics etcetera) often involves large clusters of particles which interact with external objects and each other. The particles each have individual properties (spin, momentum, direction etc) and are being created and destroyed continuously. Therefore the order of the particles is unimportant, what is important is the speed of erasure and insertion. No current standard library container has both strong insertion and non-back erasure speed, so again this is a good match for colony.

Reports from other fields suggest that, because most developers aren't aware of containers such as this, they often end up using solutions which are sub-par for iterative performance such as std::map and std::list in order to preserve pointer validity, when most of their processing work is actually iteration-based. So, introducing this container would both create a convenient solution to these situations, as well as increasing awareness of better-performing approaches in general. It will also ease communication across fields, as opposed to the current scenario where each field uses a similar container but each has a different name for it.

IV. Impact On the Standard

This is purely a library addition, requiring no changes to the language.

V. Design Decisions

The three core aspects of a colony from an abstract perspective are:

1. A collection of element memory blocks + metadata, to prevent reallocation during insertion (as opposed to a single memory block)
2. A method of skipping erased elements in O(1) time during iteration (as opposed to reallocating subsequent elements during erasure)
3. An erased-element location recording mechanism, to enable the re-use of memory from erased elements in

subsequent insertions, which in turn increases cache locality and reduces the number of block allocations/deallocations

Each memory block houses multiple elements. The metadata about each block may or may not be allocated with the blocks themselves (could be contained in a separate structure). This metadata should include at a minimum, the number of non-erased elements within each block and the block's capacity - which allows the container to know when the block is empty and needs to be removed from the iterative chain, and also allows iterators to judge when the end of one block has been reached. A non-boolean method of skipping over erased elements during iteration while maintaining O(1) amortized iteration time complexity is required (amortized due to block traversal, which would typically require a few more operations). Finally, a mechanism for keeping track of elements which have been erased must be present, so that those memory locations can be reused upon subsequent element insertions.

The following aspects of a colony must be implementation-defined in order to allow for variance and possible performance improvement, and to conform with possible changes to C++ in the future:

- the method used to skip erased elements
- time complexity of operations to update whatever metadata is associated with the skip method
- erasure-recording mechanism
- element memory block metadata
- iterator structure
- memory block growth factor
- time complexity of advance()/next()/prev()

However the implementation of these *is* significantly constrained by the requirements of the container (lack of reallocation, stable pointers to non-erased elements regardless of erasures/insertions).

In terms of the reference implementation the specific structure and mechanisms have changed many times over the course of development, however the interface to the container and its time complexity guarantees have remained largely unchanged (with the exception of the time complexity for updating skipfield nodes - which has not impacted significantly on performance). So it is reasonably likely that regardless of specific implementation, it will be possible to maintain this general specification without obviating future improvements in implementation, so long as time complexity guarantees for the above list are implementation-defined.

Below I explain the reference implementation's approach in terms of the three core aspects described above, along with descriptions of some alternatives implementation approaches.

1. Collection of element memory blocks + metadata

In the reference implementation this is essentially a doubly-linked list of 'group' structs containing (a) a dynamically-allocated element memory block, (b) memory block metadata and (c) a dynamically-allocated skipfield. The memory blocks and skipfields have a growth factor of 2 from one group to the next. The metadata includes information necessary for an iterator to iterate over colony elements, such as the last insertion point within the memory block, and other information useful to specific functions, such as the total number of non-erased elements in the node. This approach keeps the operation of freeing empty memory blocks from the colony container at O(1) time complexity. Further information is available here.

Using a vector of group structs with dynamically-allocated element memory blocks, using the swap-and-pop idiom where groups need to be erased from the iterative sequence, would not work. To explain, when a group becomes empty of elements, it must be removed from the sequence of groups, because otherwise you end up with highly-variable latency during iteration due to the need to skip over an unknown number of empty groups when traversing from one non-empty group to the next. Simply erasing the group will not suffice, as this would create a variable amount of latency during erasure when the group becomes empty, based on the number of groups after that group which would need to be reallocated backward in the vector. But even if one swapped the to-be-erased group with the back group, and then pop'd the to-be-erased group off the back, this would not solve the problem, as iterators require a stable pointer to the group they are traversing in order to traverse to the next group in the sequence. If an iterator pointed to an element in the back group, and the back group was swapped with the to-be-erased group, this would invalidate the iterator.

A vector of pointers to group structs is more-possible. Erasing groups would still have highly-variable latency due to reallocation, however the cost of reallocating pointers may be negligible depending on architecture. While the number of pointers can be expected to be low in most cases due to the growth factor in memory blocks, if the user has defined their own memory block capacity limits the number of pointers could be large, and this has to be taken into consideration. In this case using a pop-and-swap idiom is still not possible, because while it would not necessarily invalidate the internal references of an iterator pointing to an element within the back group, the sequence of blocks

would be changed and therefore the iterator would be moved backwards in the iterative sequence.

A vector of memory blocks, as opposed to a vector of pointers to memory blocks or a vector of group structs with dynamically-allocated memory blocks, would also not work, both due to the above points and because as it would (a) disallow a growth factor in the memory blocks and (b) invalidate pointers to elements in subsequent blocks when a memory block became empty of elements and was therefore removed from the vector. In short, negating colony's beneficial aspects.

2. A non-boolean method of skipping erased elements in O(1) time during iteration

The reference implementation currently uses a skipfield pattern called the Low complexity jump-counting pattern. This effectively encodes the length of runs of consecutive erased elements, into a skipfield, which allows for O(1) time complexity during iteration. Since there is no branching involved in iterating over the skipfield aside from end-of-block checks, it can be less problematic computationally than a boolean skipfield (which has to branch for every skipfield read) in terms of CPUs which don't handle branching or branch-prediction failure efficiently (eg. Core2). It also does not have the variable latency associated with a boolean skipfield.

The pattern stores and modifies the run-lengths during insertion and erasure with O(1) time complexity. It has a lot of similarities to the High complexity jump-counting pattern, which was a pattern previously used by the reference implementation. Using the High complexity jump-counting pattern is an alternative, though the skipfield update time complexity guarantees for that pattern are effectively undefined, or between O(1) and O(skipfield length) for each insertion/erasure. In practice those updates result in one memcpy operation which resolves to a single block-copy operation, but it is still a little slower than the Low complexity jump-counting pattern. The method you use to skip erased elements will typically also have an effect on the type of memory-reuse mechanism you can utilize.

A pure boolean skipfield is not usable because it makes iteration time complexity undefined - it could for example result in thousands of branching statements + skipfield reads for a single ++ operation in the case of many consecutive erased elements. In the high-performance fields for which this container was initially designed, this brings with it unacceptable latency. However another strategy using a combination of a jump-counting *and* boolean skipfield, which saves memory at the expense of computational efficiency, is possible as follows:

1. Instead of storing the data for the low complexity jump-counting pattern in it's own skipfield, have a boolean bitfield indicating which elements are erased. Store the jump-counting data in the erased element's memory space instead (possibly alongside free list data).
2. When iterating, check whether the element is erased or not using the boolean bitfield; if it is not erased, do nothing. If it is erased, read the jump value from the erased element's memory space and skip forward the appropriate number of nodes both in the element memory block and the boolean bitfield.

This approach has the advantage of still performing O(1) iterations from one non-erased element to the next, unlike a pure boolean skipfield approach, but compared to a pure jump-counting approach introduces 3 additional costs per iteration via (1) a branch operation when checking the bitfield, (2) an additional read (of the erased element's memory space) and (3) a bitmasking operation + bitshift to read the bit. But it does reduce the memory overhead of the skipfield to 1 bit per-element, which reduces the cache load. An implementation and benchmarking would be required in order to establish how this approach compares to the current implementation's performance.

Another method worth mentioning is the use of a referencing array - for example, having a vector of elements, together with a vector of either indexes or pointers to those elements. When an element is erased, the vector of elements itself is not updated - no elements are reallocated. Meanwhile the referencing vector is updated and the index or pointer to the erased element is erased. When iteration occurs it iterates over the referencing vector, accessing each element in the element vector via the indexes/pointers. The disadvantages of this technique are (a) much higher memory usage, particularly for small elements and (b) highly-variable latency during erasure due to reallocation in the referencing array. Since once of the goals of colony is predictable latency, this is likely not suitable.

Packed arrays are not worth mentioning as the iteration method is considered separate from the referencing mechanism, making them unsuitable for a std:: container.

3. Erased-element location recording mechanism

There are two valid approaches here; both involve per-memory-block free lists, utilizing the memory space of erased elements. The first approach forms a free list of all erased elements. The second forms a free list of the first element in each *run* of consecutive erased elements ("skipblocks", in terms of the terminology used in the jump-counting pattern papers). The second can be more efficient, but requires a doubly-linked free list rather than a singly-linked free list, at least with a jump-counting skipfield - otherwise it would become an O(N) operation to update links in the skipfield, when a skipblock expands or contracts during erasure or insertion.

The reference implementation currently uses the second approach, using three things to keep track of erased element

locations:

a. Metadata for each memory block includes a 'next block with erasures' pointer. The container itself contains a 'blocks with erasures' list-head pointer. These are used by the container to create an intrusive singly-linked list of memory blocks with erased elements which can be re-used for future insertions.

b. Metadata for each memory block also includes a 'free list head' index number, which records the index (within the memory block) of the first element of the last-created skipblock - the 'head' skipblock.

c. The memory space of the first erased element in each skipblock is reinterpret_cast'd via pointers as two index numbers, the first giving the index of the previous skipblock in that memory block, the second giving the index of the next skipblock in the sequence. In the case of the 'head' skipblock in the sequence, a unique number is used for the 'next' index. This forms a free list of runs of erased element memory locations which may be re-used.

Previous versions of the reference implementation used a singly-linked free list of erased elements instead of a doubly-linked free list of skipblocks, this was possible with the High complexity jump-counting pattern, but not possible using the Low complexity jump-counting pattern as it cannot calculate a skipblock's start node location from a middle node's value like the High complexity pattern can. Using free-lists of skipblocks is a more efficient approach.

One cannot use a stack of pointers (or similar) to erased elements for this mechanism, as early versions of the reference implementation did, because this can create allocations during erasure, which changes the exception guarantees of erase(). One could instead scan all skipfields until an erased location was found, or simply have the first item in the list above and then scan the first available block, though both of these approaches would be slow.

In terms of the alternative *boolean + jump-counting skipfield* approach described in the erased-element-skip-method section above, one could store both the jump-counting data and free list data in any given erased element's memory space, provided of course that elements are aligned to be wide enough to fit both.

Implementation of iterator class

Any iterator implementation is going to be dependent on the erased-element-skipping mechanism used. The reference implementation's iterator stores a pointer to the current 'group' struct mentioned above, plus a pointer to the current element and a pointer to its corresponding skipfield node. An alternative approach is to store the group pointer + an index, since the index can indicate both the offset from the memory block for the element, as well as the offset from the start of the skipfield for the skipfield node. However multiple implementations and benchmarks across many processors have shown this to be worse-performing than the separate pointer-based approach, despite the increased memory cost for the iterator class itself.

++ operation is as follows, utilising the reference implementation's Low-complexity jump-counting pattern:

1. Add 1 to the existing element and skipfield pointers.
2. Dereference skipfield pointer to get value of skipfield node, add value of skipfield node to both the skipfield pointer and the element pointer. If the node is erased, its value will be a positive integer indicating the number of nodes until the next non-erased node, if not erased it will be zero.
3. If element pointer is now beyond end of element memory block, change group pointer to next group, element pointer to the start of the next group's element memory block, skipfield pointer to the start of the next group's skipfield. In case there is a skipblock at the beginning of this memory block, dereference skipfield pointer to get value of skipfield node and add value of skipfield node to both the skipfield pointer and the element pointer. There is no need to repeat the check for end of block, as the block would have been removed from the iteration sequence if it were empty of elements.

-- operation is the same except both step 1 and 2 involve subtraction rather than adding, and step 3 checks to see if the element pointer is now before the beginning of the memory block. If so it traverses to the back of the previous group, and subtracts the value of the back skipfield node from the element pointer and skipfield pointer.

Iterators are bidirectional but also provide constant time complexity >, <, >=, <= and <=> operators for convenience (eg. in `for` loops when skipping over multiple elements per loop and there is a possibility of going past a pre-determined end element). This is achieved by keeping a record of the order of memory blocks. In the reference implementation this is done by assigning a number to each memory block in its metadata. In an implementation using a vector of pointers to memory blocks instead of a linked list, one could use the position of the pointers within the vector to determine this. Comparing relative order of the two iterators' memory blocks via this number, then comparing the memory locations of the elements themselves, if they happen to be in the same memory block, is enough to implement all greater/lesser comparisons.

Additional notes on specific functions

- **`iterator insert`** (all variants)

  Insertion re-uses previously-erased element memory locations when available, so position of insertion is effectively random unless no previous erasures have occurred, in which case all elements will be inserted linearly to the back of the container, at least in the current implementation. These details have been removed from the standard in order to allow leeway for potentially-better implementations in future - though it is expected that a colony will always reuse erased memory locations, it is impossible to predict optimal strategies for unknown future hardware.

- **`void insert`** (all variants)

  For range, fill and initializer_list insertion, it is not possible to guarantee that all the elements inserted will be sequential in the colony's iterative sequence, and therefore it is not considered useful to return an iterator to the first inserted element. There is a precedent for this in the various std:: map containers. Therefore these functions return void presently.

  For range insert and range constructors, thhe syntax has been modified compared to other containers in order to take two potentially-different iterator types in order to support sentinels and the like.

- **`iterator erase`** (all variants)

  Firstly it should be noted that erase may retain memory blocks which become completely empty of elements due to erasures, adding them to the set of unused memory blocks which are normally created by reserve(). Under what circumstances these memory blocks are retained rather than deallocated is implementation-defined - however given that small memory blocks have low cache locality compared to larger ones, from a performance perspective it is best to only retain the larger of the blocks currently allocated in the colony. In most cases this would mean the back block would almost always be retained. There is a lot of nuance to this, and it's also a matter of trading off complexity of implementation vs actual benchmarked speed vs latency. In my tests retaining both back blocks and 2nd-to-back blocks while ignoring actual capacity of blocks seems to have the best overall performance characteristics.

  There are three major performance advantages to retaining back blocks as opposed to any block - the first is that these will be, under most circumstances, the largest blocks in the colony (given the built-in growth factor) - the only exception to this is when splice is used, which may result in a smaller block following a larger block (implementation-dependent). Larger blocks == more cache locality during iteration. The second advantage is that in situations where elements are being inserted to and erased from the back of the colony (this assumes no erased element locations in other memory blocks, which would otherwise be used for insertions) continuously and in quick succession, retaining the back block avoids large numbers of deallocations/reallocations. The third advantage is that deallocations of larger blocks can, in part, be moved to non-critical code regions via trim(). Though ultimately if the user wants total control of when allocations and deallocations occur they would want to use a custom allocator.

  Lastly, specifying a return iterator for range-erase may seem pointless, as no reallocation of elements occurs in erase so the return iterator will almost always be the `last` iterator of the `const_iterator first, const_iterator last` pair. However if `last` was `end()`, the new value of `end()` (if it has changed due to empty block removal) will be returned. In this case either the user submitted `end()` as `last`, or they incremented an iterator pointing to the final element in the colony and submitted that as `last`. The latter is the only valid reason to return an iterator from the function, as it may occur as part of a loop which is erasing elements and ends when `end()` is reached. If `end()` is changed by the erasure of an entire memory block, but the iterator being used in the loop does not accurately reflect `end()`'s new value, that iterator could iterate past `end()` and the loop would never finish.

- **`void reshape(std::colony_limits block_capacity_limits);`**

  The order of elements post-reshape is not guaranteed to be stable in order to allow for optimizations. Specifically in the instance where a given element memory block no longer fits within the limits supplied by the user, depending on the state of the colony as a whole, the elements within that memory block could be reallocated to previously-erased element locations in other memory blocks which do fit within the supplied limits.

  Additionally if there is empty capacity at the back of the last block in the container, at least some of the elements could be moved to that position rather than being reallocated to a new memory block. Both of these techniques increase cache locality by removing skipped memory spaces within existing memory blocks. However whether they are used is implementation-dependent.

- **`iterator get_iterator_from_pointer(pointer p) const noexcept;`**

  Because colony iterators are likely to be large, storing three pieces of data - current memory block, current element within memory block and potentially, current skipfield node - a program storing many links to elements within a colony may opt to dereference iterators to get pointers and store those instead of iterators, to save memory. This function reverses the process, giving an iterator which can then be used for operations such as erase.

- **`void shrink_to_fit();`**

A decision had to be made as to whether this function should, in the context of colony, be allowed to reallocate elements (as std::vector does) or simply trim off unused memory blocks (as std::deque does). Due to the fact that a large colony memory block could have as few as one remaining element after a series of erasures, it makes little sense to only trim unused blocks, and instead a shrink_to_fit is expected to reallocate all non-erased elements to as few memory blocks as possible in order to increase cache locality during iteration and reduce memory use. As with reshape(), the order of elements post-reshape is not guaranteed to be stable, to allow for potential optimizations. The trim() command is also introduced as a way to free unused memory blocks which have been previously reserved, without reallocating elements and invalidating iterators.

- **`void sort();`**

  It is forseen that although the container has unordered insertion, there may be circumstances within which sorting is desired. Because colony uses bidirectional iterators, using std::sort or similar is not possible. Therefore an internal sort routine is warranted, as it is with std::list. An implementation of the sort routine used in the reference implementation of colony can be found in a non-container-specific form at plflib.org/indiesort.htm - see that page for the technique's advantages over the usual sort algorithms for non-random-access containers. Unfortunately to date there has been no interest in including this algorithm in the standard library. An allowance is made for sort to allocate memory if necessary, so that algorithms such as indiesort can be used internally.

- **`void splice(colony &x);`**

  Whether `x`'s blocks are transferred to the beginning or end of `*this`'s iterative sequence, or interlaced in some way (for example, to preserve relative capacity growth-factor ordering of subsequent blocks) is implementation-defined. Better performance may be gained in some cases by allowing the source blocks to go to the front rather than the back, depending on how full the final block in `x`'s iterative sequence is. This is because unused elements that are not at the back of colony's iterative sequence will need to be marked as skipped, and skipping over large numbers of elements will incur a small performance disadvantage during iteration compared to skipping over a small number of elements, due to memory locality.

  In addition, whether this function is `noexcept` is implementation-defined - in the case of an implementation using a linked list of groups (like the reference implementation) this function can be `noexcept(allocator_traits<Allocator>::is_always_equal::value)`. However, in the case of an implementation using a vector of pointers to groups, an additional allocation may have to be made if the group pointer vector isn't of sufficient capacity to accomodate the spliced groups from source `x` - which could potentially trigger an exception.

- **`size_type memory() const noexcept;`**

  A colony uses memory block metadata and may use a skipfield, both which are implementation-defined, so it is not possible for a user to estimate internal memory usage from size(), sizeof() or capacity(). This function fulfills that role. Because some types of elements may allocate their own memory dynamically (eg. std::colony<std::vector>) only the static allocation of each element is included in this functions byte count.

  This function can be made constant time by adding a counter to the colony that keeps track of the number of reserved memory blocks available, or by having a vector of pointers to memory blocks instead an intrusive linked list of memory blocks. However in the case of the reference implementation which uses linked lists, the counter metadata would only be used by this function and since this function is not expected to be in heavy use, the time complexity of this function is left as implementation-defined to enable best performance.

- Non-member function overloads for **`advance, prev and next`** (all variants)

  For these functions, complexity is dependent on state of colony, position of iterator and amount of distance, but in many cases will be less than linear, and may be constant. To explain: it is necessary in a colony to store metadata both about the capacity of each block (for the purpose of iteration) and how many non-erased elements are present within the block (for the purpose of removing blocks from the iterative chain once they become empty). For this reason, intermediary blocks between the iterator's initial block and its final destination block (if these are not the same block, and if the initial block and final block are not immediately adjacent) can be skipped rather than iterated linearly across, by using the "number of non-erased elements" metadata.

  This means that the only linear time operations are any iterations within the initial block and the final block. However if either the initial or final block have no erased elements (as determined by comparing whether the block's capacity metadata and the block's "number of non-erased elements" metadata are equal), linear iteration can be skipped for that block and pointer/index math used instead to determine distances, reducing complexity to constant time. Hence the best case for this operation is constant time, the worst is linear to the distance.

- Non-member function overloads for **`distance`** (all variants)

  The same considerations which apply to advance, prev and next also apply to distance - intermediary blocks between first and last's blocks can be skipped in constant time and their "number of non-erased elements" metadata added to the cumulative distance count, while first's block and last's block (if they are not the same block) must be linearly iterated across unless either block has no erased elements, in which case the operation becomes pointer/index math and is reduced to constant time for that block. In addition, if first's block is not the

same as last's block, and last is equal to end() or --end(), or is the last element in that block, last's block's elements can also counted from the "number of non-erased elements" metadata rather than via iteration.

- Priority template parameter for the container

  This forms a non-binding request for the container to prioritize either performance or memory use, supplied in the form of a scoped enum. The reference implementation uses a regular un-scoped enum, as it must also work under C++03. In terms of the reference implementation the priority parameter changes the skipfield type from unsigned short (performance) to unsigned char (memory use) - which in turn changes the maximum block limits, because in the reference implementation the block capacities are limited to numeric_limits<skipfield_type>::max. The maximum block capacity limit affects iteration performance, due to a greater or lesser number of elements being able to be sequential in memory, and the subsequent effects on cache. For small numbers of elements ie. under 1000, unsigned char also will be faster in addition to needing less memory, due to the lowered cache usage and the fact that the maximum block capacity limit is no significantly limiting cache locality at this point. Hence, prioritizing for performance may not necessarily be faster in all circumstances, but should be faster in most - if in fact the request is actioned upon, and it is not guaranteed to be actioned upon in all implementations.

  There is a point of diminishing returns in terms of how many elements can be stored sequentially in memory and how that impacts performance, due to the limits of cache size - hence it was found that increasing the skipfield type to unsigned int and thence increasing the block capacity limit, did not have a performance advantage on any number of elements.

Results of implementation

In practical application the reference implementation is generally faster for insertion and (non-back) erasure than current standard library containers, and generally faster for iteration than any container except vector and deque. For full details, see benchmarks.

VI. Technical Specification

Suggested location of colony in the standard is 22.3, Sequence Containers.

22.3.7 Header <colony> synopsis [colony.syn]

```
#include <initializer_list> // see 17.10.2
#include <compare> // see 17.11.1
#include <concepts> // see 18.3
#include <stdexcept> // see 19.2
#include <utility> // see 20.2.1
#include <memory> // see 20.10

namespace std {
   // 22.3.14, class template colony

        struct colony_limits;
        enum class colony_priority;

   template <class T, class Allocator = allocator<T>, colony_priority priority =
colony_priority::performance> class colony;

   namespace pmr {
      template <class T>
      using colony = std::colony<T, polymorphic_allocator<T>>;
   }
}
```

Iterator Invalidation

All read-only operations, swap, std::swap, splice, operator= && (source), reserve, trim	Never.
clear, operator= & (destination), operator= && (destination)	Always.
reshape	Only if memory blocks exist whose capacities do not fit within the supplied limits.
shrink_to_fit	Only if capacity() != size().
erase	Only for the erased element. If an iterator is == end() it may be invalidated if the back element of the colony is erased (similar to deque (22.3.9)). Likewise if a reverse_iterator is == rend() it may be invalidated if the front element of the colony is erased. The same applies with cend() and crend() for const_iterator and const_reverse_iterator respectively.
insert, emplace	If an iterator is == end() or == begin() it may be invalidated by a subsequent insert/emplace. Likewise if a reverse_iterator is == rend() or == rbegin() it may be invalidated by a subsequent insert/emplace. The same rules apply with cend(), cbegin() and crend(), crbegin() for const_iterator and const_reverse_iterator respectively.

22.3.14 Class template `colony` [colony]

22.3.14.1 Class template `colony` overview [colony.overview]

1. A colony is a sequence container that allows constant-time insert and erase operations. Insertion location is the back of the container when no erasures have occured. When erasures have occured it will re-use existing erased element memory spaces where possible and insert to those locations. Storage management is handled automatically and is specifically organized in multiple blocks of sequential elements. Unlike vectors (22.3.12) and deques (22.3.9), fast random access to colony elements is not supported, but specializations of advance/next/prev give access which can be better than linear time in the number of elements traversed.
2. Erasures are processed using implementation-defined strategies for skipping erased elements during iteration, rather than reallocating subsequent elements as is expected in a vector or deque.
3. Memory block element capacities have an implementation-defined growth factor, for example each block can be twice the capacity of the preceding block.
4. Limits can be placed on the minimum and maximum element capacities of memory blocks, both by a user and by an implementation. Minimum capacity shall be nor more than maximum capacity. When limits are not specified by a user, the implementation's limits are used. Where user-specified limits do not fit within the implementation's limits (ie. user minimum is less than implementation minimum or user maximum is more than implementation maximum) an exception is thrown. User-specified limits can be supplied to a constructor or to the reshape() function, using the `std::colony_limits` struct with its `min` and `max` members set to the minimum and maximum element capacity limits respectively. The current limits in a colony instance can be obtained from block_capacity_limits().
5. A colony satisfies all of the requirements of a container, of a reversible container (given in two tables in 22.2), of a sequence container, including most of the optional sequence container requirements (22.2.3), and of an allocator-aware container (Table 78). The exceptions are the `operator[]` and `at` member functions, which are not provided.
6. Colony iterators satisfy bidirectional requirements but also provide relational operators <, <=, >, >= and <=> which compare the relative ordering of two iterators in the sequence of a colony instance.
7. Iterator operations ++ and -- take constant amortized time, other iterator operations take constant time.

```
template <class T, class Allocator = std::allocator<T>, priority Priority = priority::performance> class colony
```

`T` - the element type. In general T shall meet the requirements of Erasable, CopyAssignable and CopyConstructible. However, if emplace is utilized to insert elements into the colony, and no functions which involve copying or moving are utilized, T is only required to meet the requirements of Erasable.
If move-insert is utilized instead of emplace, T shall also meet the requirements of MoveConstructible.

`Allocator` - an allocator that is used to acquire memory to store the elements. The type shall meet the requirements of Allocator. The behavior is undefined if `Allocator::value_type` is not the same as T.

`Priority` - if set to `priority::memory_use` this is a non-binding request to prioritize lowered memory usage over container performance. [Note: The request is non-binding to allow latitude for implementation-specific optimizations. If this feature is implemented, it is *not* specified that the container shall have better performance when using priority::performance instead of priority::memory_usage in *all* scenarios, but that it shall have better performance in *most* scenarios. — end note]

```
namespace std {

struct colony_limits
{
        size_t min, max;
        colony_limits(const size_t minimum, const size_t maximum) noexcept : min(minimum),
max(maximum) {}
};


enum struct colony_priority { performance, memory_use };


template <class T, class Allocator = allocator<T>, colony_priority Priority =
colony_priority::performance>
class colony {
public:

  // types
  using value_type = T;
  using allocator_type = Allocator;
  using pointer = typename allocator_traits<Allocator>::pointer;
  using const_pointer = typename allocator_traits<Allocator>::const_pointer;
  using reference = value_type&;
  using const_reference = const value_type&;
  using size_type = implementation-defined; // see 22.2
  using difference_type = implementation-defined; // see 22.2
  using iterator = implementation-defined; // see 22.2
  using const_iterator = implementation-defined; // see 22.2
  using reverse_iterator = implementation-defined; // see 22.2
  using const_reverse_iterator = implementation-defined; // see 22.2



  colony() noexcept(noexcept(Allocator())) : colony(Allocator()) { }
  explicit colony(std::colony_limits block_capacity_limits)
noexcept(noexcept(Allocator())) : colony(Allocator()) { }
  explicit colony(const Allocator&) noexcept;
  explicit colony(std::colony_limits block_capacity_limits, const Allocator&) noexcept;
  explicit colony(size_type n, std::colony_limits block_capacity_limits = implementation-
defined, const Allocator& = Allocator());
  colony(size_type n, const T& value, std::colony_limits block_capacity_limits =
implementation-defined, const Allocator& = Allocator());
  template <class InputIterator>
    colony(InputIterator first, InputIterator last, std::colony_limits
block_capacity_limits = implementation-defined, const Allocator& = Allocator());
  colony(const colony& x);
  colony(colony&&) noexcept;
  colony(const colony&, const Allocator&);
  colony(colony&&, const Allocator&);
  colony(initializer_list<T>, std::colony_limits block_capacity_limits = implementation-
defined, const Allocator& = Allocator());
  ~colony() noexcept;
  colony& operator= (const colony& x);
  colony& operator= (colony&& x)
noexcept(allocator_traits<Allocator>::propagate_on_container_move_assignment::value ||
allocator_traits<Allocator>::is_always_equal::value);
  colony& operator= (initializer_list<T>);
  template<class InputIterator> void assign(InputIterator first, InputIterator last);
  void assign(size_type n, const T& t);
  void assign(initializer_list<T>);
```

```
  allocator_type get_allocator() const noexcept;



  // iterators
  iterator               begin() noexcept;
  const_iterator         begin() const noexcept;
  iterator               end() noexcept;
  const_iterator         end() const noexcept;
  reverse_iterator       rbegin() noexcept;
  const_reverse_iterator rbegin() const noexcept;
  reverse_iterator       rend() noexcept;
  const_reverse_iterator rend() const noexcept;

  const_iterator         cbegin() const noexcept;
  const_iterator         cend() const noexcept;
  const_reverse_iterator crbegin() const noexcept;
  const_reverse_iterator crend() const noexcept;


  // capacity
  [[nodiscard]] bool empty() const noexcept;
  size_type size() const noexcept;
  size_type max_size() const noexcept;
  size_type capacity() const noexcept;
  size_type memory() const noexcept;
  void reserve(size_type n);
  void shrink_to_fit();
  void trim() noexcept;


  // modifiers
  template <class... Args> iterator emplace(Args&&... args);
  iterator insert(const T& x);
  iterator insert(T&& x);
  void insert(size_type n, const T& x);
  template <class InputIterator1, class InputIterator2> void insert(InputIterator first,
InputIterator2 last);
  void insert(initializer_list<T> il);
  iterator erase(const_iterator position);
  iterator erase(const_iterator first, const_iterator last);
  void swap(colony&)
noexcept(allocator_traits<Allocator>::propagate_on_container_swap::value ||
allocator_traits<Allocator>::is_always_equal::value);
  void clear() noexcept;


  // colony operations
  void splice(colony &x);

  std::colony_limits block_capacity_limits() const noexcept;
  void reshape(std::colony_limits block_capacity_limits);

  iterator get_iterator_from_pointer(pointer p) const noexcept;

  void sort();
  template <class Compare> void sort(Compare comp);

  friend bool operator== (const colony &x, const colony &y);
  friend bool operator!= (const colony &x, const colony &y);


  class iterator
  {
    friend void advance(iterator &it, Distance n);
    friend iterator next(const iterator it, difference_type distance = 1);
    friend iterator prev(const iterator it, difference_type distance = 1);
    friend difference_type distance(const iterator first, const iterator last);
  }
```

```
  class const_iterator
  {
    friend void advance(const_iterator &it, Distance n);
    friend const_iterator next(const const_iterator it, difference_type distance = 1);
    friend const_iterator prev(const const_iterator it, difference_type distance = 1);
    friend difference_type distance(const const_iterator first, const const_iterator
last);
  }


  class reverse_iterator
  {
    friend void advance(reverse_iterator &it, Distance n);
    friend reverse_iterator next(const reverse_iterator it, difference_type distance = 1);
    friend reverse_iterator prev(const reverse_iterator it, difference_type distance = 1);
    friend difference_type distance(const reverse_iterator first, const reverse_iterator
last);
  }


  class const_reverse_iterator
  {
    friend void advance(const_reverse_iterator &it, Distance n);
    friend const_reverse_iterator next(const const_reverse_iterator it, difference_type
distance = 1);
    friend const_reverse_iterator prev(const const_reverse_iterator it, difference_type
distance = 1);
    friend difference_type distance(const const_reverse_iterator first, const
const_reverse_iterator last);
  }


  // swap
  friend void swap(colony& x, colony& y)
    noexcept(noexcept(x.swap(y)));


  // erase
  template <class Predicate>
    friend size_type erase_if(colony& c, Predicate pred);
  template <class U>
    friend size_type erase(colony& c, const U& value);
}


template<class InputIterator, class Allocator = allocator<iter-value-type
<InputIterator>>>
  colony(InputIterator, InputIterator, Allocator = Allocator())
    -> colony<iter-value-type <InputIterator>, Allocator>;
```

22.3.14.2 colony constructors, copy, and assignment [colony.cons]

`explicit colony(const Allocator&);`

1. Effects: Constructs an empty colony, using the specified allocator.
2. Complexity: Constant.

`explicit colony(size_type n, const T& value, std::colony_limits block_capacities = implementation-defined, const Allocator& =Allocator());`

3. Preconditions: `T` shall be *Cpp17MoveInsertable* into `*this`.
4. Effects: Constructs a colony with n copies of `value`, using the specified allocator.
5. Complexity: Linear in n.
6. Throws: `length_error` if `block_capacities.min` or `block_capacities.max` are outside the implementation's

minimum and maximum element memory block capacity limits, or if `block_capacities.min > block_capacities.max`.

7. Remarks: If `n` is larger than `block_capacities.min`, the capacity of the first block created will be the smaller of `n` or `block_capacities.max`.

```
template <class InputIterator1, class InputIterator2>
  colony(InputIterator1 first, InputIterator2 last, std::colony_limits block_capacities = implementation-defined, const Allocator& = Allocator());
```

8. Preconditions: `InputIterator1` shall be `std::equality_comparable_with InputIterator2`.
9. Effects: Constructs a colony equal to the range [first, last), using the specified allocator.
10. Complexity: Linear in distance(first, last).
11. Throws: `length_error` if `block_capacities.min` or `block_capacities.max` are outside the implementation's minimum and maximum element memory block capacity limits, or if `block_capacities.min > block_capacities.max`. Or
12. Remarks: If iterators are random-access, let `n` be last - first; if `n` is larger than `block_capacities.min`, the capacity of the first block created will be the smaller of `n` or `block_capacities.max`.

22.3.14.3 colony capacity [colony.capacity]

```
size_type capacity() const noexcept;
```

1. Returns: The total number of elements that the colony can currently contain without needing to allocate more memory blocks.

```
size_type memory() const noexcept;
```

2. Returns: The memory use, in bytes, of the container as a whole, including elements but not including any dynamic allocation incurred by those elements.

```
void reserve(size_type n);
```

3. Effects: A directive that informs a colony of a planned change in size, so that it can manage the storage allocation accordingly. Since minimum and maximum memory block sizes can be specified by users, after `reserve()`, `capacity()` is not guaranteed to be equal to the argument of `reserve()`, may be greater. Does not cause reallocation of elements.
4. Complexity: It does not change the size of the sequence and creates at most `(n / block_capacity_limits().max) + 1` allocations.
5. Throws: `length_error` if `n > max_size()`[223].

223) reserve() uses Allocator::allocate() which may throw an appropriate exception.

```
void shrink_to_fit();
```

6. Preconditions: `T` is *Cpp17MoveInsertable* into `*this`.
7. Effects: shrink_to_fit is a non-binding request to reduce `capacity()` to be closer to `size()`. [Note: The request is non-binding to allow latitude for implementation-specific optimizations. — end note] It does not increase `capacity()`, but may reduce `capacity()` by causing reallocation. It may move elements from multiple memory blocks and consolidate them into a smaller number of memory blocks.
   If an exception is thrown other than by the move constructor of a non-*Cpp17CopyInsertable* T, there are no effects.
8. Complexity: If reallocation happens, linear to the number of elements reallocated.
9. Remarks: Reallocation invalidates all the references, pointers, and iterators referring to the elements reallocated as well as the past-the-end iterator. [Note: If no reallocation happens, they remain valid. —end note] The order of elements post-operation is not guaranteed to be stable.

```
void trim();
```

10. Effects: Removes and deallocates empty memory blocks created by prior calls to `reserve()` or `erase()`. If such memory blocks are present, `capacity()` will be reduced.
11. Complexity: Linear in the number of reserved blocks to deallocate.
12. Remarks: Does not reallocate elements and no references, pointers or iterators referring to elements in the sequence will be invalidated.

22.3.14.4 colony modifiers [colony.modifiers]

```
iterator insert(const T& x);
iterator insert(T&& x);
void insert(size_type n, const T& x);
template <class InputIterator1, class InputIterator2>
  void insert(InputIterator1 first, InputIterator2 last);
void insert(initializer_list<T>);
template <class... Args>
  iterator emplace(Args&&... args);
```

1. Preconditions: For `template <class InputIterator1, class InputIterator2> void insert(InputIterator1 first, InputIterator2 last)`, `InputIterator1` shall be `std::equality_comparable_with InputIterator2`.
2. Complexity: Insertion of a single element into a colony takes constant time and exactly one call to a constructor of `T`. Insertion of multiple elements into a colony is linear in the number of elements inserted, and the number of calls to the copy constructor or move constructor of `T` is exactly equal to the number of elements inserted.
3. Remarks: Does not affect the validity of iterators and references, unless an iterator points to `end()`, in which case it may be invalidated. Likewise if a reverse_iterator points to `rend()` it may be invalidated. If an exception is thrown there are no effects.

```
iterator erase(const_iterator position);
```

3. Effects: Invalidates only the iterators and references to the erased element.
4. Complexity: Constant. [Note: operations pertaining to the updating of any data associated with the erased-elemment skipping mechanism is not factored into this; it is implementation-defined and may be constant, linear or otherwise defined. —end note]

```
iterator erase(const_iterator first, const_iterator last);
```

5. Effects: Invalidates only the iterators and references to the erased elements. In some cases if an iterator is equal to `end()` and the back element of the colony is erased, that iterator may be invalidated. Likewise if a reverse_iterator is equal to `rend()` and the front element of the colony is erased, that reverse_iterator may be invalidated.
6. Complexity: Linear in the number of elements erased for non-trivially-destructible types, for trivially-destructible types constant in best case and linear in worst case, approximating logarithmic in the number of elements erased on average.

```
void swap(colony& x) noexcept(allocator_traits<Allocator>::propagate_on_container_swap::value ||
allocator_traits<Allocator>::is_always_equal::value);
```

7. Effects: Exchanges the contents and `capacity()` of `*this` with that of `x`.
8. Complexity: Constant time.

22.3.14.5 Operations [colony.operations]

```
void splice(colony &x);
```

1. Preconditions: &x != this.
2. Effects: Inserts the contents of `x` into `*this` and `x` becomes empty. Pointers and references to the moved elements of

`x` now refer to those same elements but as members of `*this`. Iterators referring to the moved elements will continue to refer to their elements, but they now behave as iterators into `*this`, not into `x`.

3. Complexity: Constant time.

```
std::colony_limits block_capacity_limits() const noexcept;
```

4. Effects: Returns a std::colony_limits struct with the `min` and `max` members set to the current minimum and maximum element memory block capacity values of `*this`.

5. Complexity: Constant time.

```
void reshape(std::colony_limits block_capacity_limits);
```

6. Preconditions: `T` shall be *Cpp17MoveInsertable* into `*this`.

7. Effects: Sets minimum and maximum element memory block capacities to the min and max members of the supplied std::colony_limits struct. If the colony is not empty, adjusts existing memory block capacities to conform to the new minimum and maximum block capacities, where necessary. If existing memory block capacities are within the supplied minimum/maximum range, no reallocation of elements takes place. If they are not within the supplied range, elements are reallocated to new memory blocks which fit within the supplied range and the old memory blocks are deallocated. Order of elements is not guaranteed to be stable.

8. Complexity: If no reallocation occurs, constant time. If reallocation occurs, complexity is linear in the number of elements reallocated.

9. Throws: `length_error` if `block_capacities.min` or `block_capacities.max` are outside the implementation's minimum and maximum element memory block capacity limits, or if `block_capacities.min > block_capacities.max`.[223]

10. Remarks: The order of elements post-operation is not guaranteed to be stable (16.5.5.8).

```
iterator get_iterator_from_pointer(pointer p) const noexcept;
```

11. Effects: Returns an iterator pointing to the same element as the pointer. If `p` does not point to an element in `*this`, `end()` is returned.

```
void sort();
template <class Compare>
  void sort(Compare comp);
```

12. Preconditions: `T` is *Cpp17MoveInsertable* into `*this`.

13. Effects: Sorts the colony according to the `operator <` or a `Compare` function object. If an exception is thrown, the order of the elements in `*this` is unspecified. Iterators and references may be invalidated.

14. Complexity: Approximately N log N comparisons, where `N == size()`.

15. Throws: `bad_alloc` if it fails to allocate any memory necessary for the sort process.

16. Remarks: Not required to be stable (16.5.5.8). May allocate memory.

22.3.14.6 Specialized algorithms [colony.special]

```
friend void swap(colony &x, colony &y) noexcept(noexcept(x.swap(y)));
```

1. Effects: As if by `x.swap(y)`.

2. Remarks: This function is to be found via argument-dependent lookup only.

```
friend bool operator== (const colony &x, const colony &y);
friend bool operator!= (const colony &x, const colony &y);
```

3. Returns: For `==`, returns `True` if both containers have the same elements in the same iterative sequence, otherwise `False`. For `!=`, returns `True` if both containers do not have the same elements in the same iterative sequence,

otherwise `False`.

4. Remarks: These functions are to be found via argument-dependent lookup only.

```
class iterator
{
  friend void advance(iterator &it, Distance n);
  friend iterator next(const iterator it, difference_type distance = 1);
  friend iterator prev(const iterator it, difference_type distance = 1);
  friend difference_type distance(const iterator first, const iterator last);
}
```

```
class const_iterator
{
  friend void advance(const_iterator &it, Distance n);
  friend const_iterator next(const const_iterator it, difference_type distance = 1);
  friend const_iterator prev(const const_iterator it, difference_type distance = 1);
  friend difference_type distance(const const_iterator first, const const_iterator last);
}
```

```
class reverse_iterator
{
  friend void advance(reverse_iterator &it, Distance n);
  friend reverse_iterator next(const reverse_iterator it, difference_type distance = 1);
  friend reverse_iterator prev(const reverse_iterator it, difference_type distance = 1);
  friend difference_type distance(const reverse_iterator first, const reverse_iterator last);
}
```

```
class const_reverse_iterator
{
  friend void advance(const_reverse_iterator &it, Distance n);
  friend const_reverse_iterator next(const const_reverse_iterator it, difference_type distance = 1);
  friend const_reverse_iterator prev(const const_reverse_iterator it, difference_type distance = 1);
  friend difference_type distance(const const_reverse_iterator first, const const_reverse_iterator last);
}
```

5. Complexity: Constant in best case and linear in the number of elements traversed in worst case, approximating logarithmic in the number of elements traversed on average.

6. Remarks: These functions are to be found via argument-dependent lookup only.

22.3.14.7 Erasure [colony.erasure]

```
template <class U>
    friend size_type erase(colony& c, const U& value);
```

1. Effects: All elements in the container which are equal to `value` are erased. Invalidates all references and iterators to the erased elements.

2. Remarks: This function is to be found via argument-dependent lookup only.

```
template <class Predicate>
    friend size_type erase_if(colony& c, Predicate pred);
```

3. Effects: All elements in the container which match predicate `pred` are erased. Invalidates all references and iterators to the erased elements.

4. Remarks: This function is to be found via argument-dependent lookup only.

VII. Acknowledgements

Matt would like to thank: Glen Fernandes and Ion Gaztanaga for restructuring advice, Robert Ramey for documentation advice, various Boost and SG14 members for support, critiques and corrections, Baptiste Wicht for teaching me how to construct decent benchmarks, Jonathan Wakely, Sean Middleditch, Jens Maurer (very nearly a co-author at this point really), Patrice Roy and Guy Davidson for standards-compliance advice and critiques, support, representation at meetings and bug reports, Henry Miller for getting me to clarify why the instrusive list/free list approach to memory location reuse is the most appropriate, that ex-Lionhead guy for annoying me enough to force me to implement the original skipfield pattern, Jon Blow for some initial advice and Mike Acton for some influence, the community at large for giving me feedback and bug reports on the reference implementation.
Also Nico Josuttis for doing such a great job in terms of explaining the general format of the structure to the committee.

VIII. Appendices

Appendix A - Basic usage examples

Using reference implementation.

```
#include <iostream>
#include <numeric>
#include "plf_colony.h"

int main(int argc, char **argv)
{
  plf::colony<int> i_colony;

  // Insert 100 ints:
  for (int i = 0; i != 100; ++i)
  {
    i_colony.insert(i);
  }

  // Erase half of them:
  for (plf::colony<int>::iterator it = i_colony.begin(); it != i_colony.end(); ++it)
  {
    it = i_colony.erase(it);
  }

  std::cout << "Total: " << std::accumulate(i_colony.begin(), i_colony.end(), 0) << std::endl;
  std::cin.get();
  return 0;
}
```

Example demonstrating pointer stability

```
#include <iostream>
#include "plf_colony.h"

int main(int argc, char **argv)
{
  plf::colony<int> i_colony;
  plf::colony<int>::iterator it;
  plf::colony<int *> p_colony;
  plf::colony<int *>::iterator p_it;

  // Insert 100 ints to i_colony and pointers to those ints to p_colony:
  for (int i = 0; i != 100; ++i)
  {
    it = i_colony.insert(i);
    p_colony.insert(&(*it));
  }

  // Erase half of the ints:
```

```
  for (it = i_colony.begin(); it != i_colony.end(); ++it)
  {
    it = i_colony.erase(it);
  }

  // Erase half of the int pointers:
  for (p_it = p_colony.begin(); p_it != p_colony.end(); ++p_it)
  {
    p_it = p_colony.erase(p_it);
  }

  // Total the remaining ints via the pointer colony (pointers will still be valid even after
insertions and erasures):
  int total = 0;

  for (p_it = p_colony.begin(); p_it != p_colony.end(); ++p_it)
  {
    total += *(*p_it);
  }

  std::cout << "Total: " << total << std::endl;

  if (total == 2500)
  {
    std::cout << "Pointers still valid!" << std::endl;
  }

  std::cin.get();
  return 0;
}
```

Appendix B - Reference implementation benchmarks

Benchmark results for the colony reference implementation under GCC on an Intel Xeon E3-1241 (Haswell) are here.

Old benchmark results for an earlier version of colony under MSVC 2015 update 3, on an Intel Xeon E3-1241 (Haswell) are here. There is no commentary for the MSVC results.

Appendix C - Frequently Asked Questions

1. **Where is it worth using a colony in place of other std:: containers?**

   As mentioned, it is worthwhile for performance reasons in situations where the order of container elements is not important and:

   a. Insertion order is unimportant

   b. Insertions and erasures to the container occur frequently in performance-critical code, ***and***

   c. Links to non-erased container elements may not be invalidated by insertion or erasure.

   Under these circumstances a colony will generally out-perform other std:: containers. In addition, because it never invalidates pointer references to container elements (except when the element being pointed to has been previously erased) it may make many programming tasks involving inter-relating structures in an object-oriented or modular environment much faster, and could be considered in those circumstances.

2. **What are some examples of situations where a colony might improve performance?**

   Some ideal situations to use a colony: cellular/atomic simulation, persistent octtrees/quadtrees, game entities or destructible-objects in a video game, particle physics, anywhere where objects are being created and destroyed continuously. Also, anywhere where a vector of pointers to dynamically-allocated objects or a std::list would typically end up being used in order to preserve pointer stability but where order is unimportant.

3. **Is it similar to a deque?**

   A deque is reasonably dissimilar to a colony - being a double-ended queue, it requires a different internal framework. In addition, being a random-access container, having a growth factor for memory blocks in a deque is problematic (though not impossible). A deque and colony have no comparable performance characteristics except for insertion (assuming a good deque implementation). Deque erasure performance varies wildly depending on the implementation, but is generally similar to vector erasure performance. A deque invalidates pointers to subsequent container elements when erasing elements, which a colony does not, and guarantees ordered insertion.

4. **What are the thread-safe guarantees?**

   Unlike a std::vector, a colony can be read from and inserted into at the same time (assuming different locations for read and write), however it cannot be iterated over and written to at the same time. If we look at a (non-concurrent implementation of) std::vector's thread-safe matrix to see which basic operations can occur at the same time, it reads as follows (please note push_back() is the same as insertion in this regard):

std::vector	Insertion	Erasure	Iteration	Read
Insertion	No	No	No	No
Erasure	No	No	No	No
Iteration	No	No	Yes	Yes
Read	No	No	Yes	Yes

   In other words, multiple reads and iterations over iterators can happen simultaneously, but the potential reallocation and pointer/iterator invalidation caused by insertion/push_back and erasure means those operations cannot occur at the same time as anything else.

   Colony on the other hand does not invalidate pointers/iterators to non-erased elements during insertion and erasure, resulting in the following matrix:

colony	Insertion	Erasure	Iteration	Read
Insertion	No	No	No	Yes
Erasure	No	No	No	Mostly*
Iteration	No	No	Yes	Yes
Read	Yes	Mostly*	Yes	Yes

   * Erasures will not invalidate iterators unless the iterator points to the erased element.

   In other words, reads may occur at the same time as insertions and erasures (provided that the element being erased is not the element being read), multiple reads and iterations may occur at the same time, but iterations may not occur at the same time as an erasure or insertion, as either of these may change the state of the skipfield which is being iterated over, if a skipfield is used in the implementation. Note that iterators pointing to end() may be invalidated by insertion.

   So, colony could be considered more inherently thread-safe than a (non-concurrent implementation of) std::vector, but still has some areas which would require mutexes or atomics to navigate in a multithreaded environment.

5. **Any pitfalls to watch out for?**

   Because erased-element memory locations may be reused by `insert()` and `emplace()`, insertion position is essentially random unless no erasures have been made, or an equal number of erasures and insertions have been made.

6. **What is the purpose of limiting memory block minimum and maximum sizes?**

One reason might be to ensure that memory blocks match a certain processor's cache or memory pathway sizes. Another reason to do this is that it is slightly slower to obtain an erased-element location from the list of groups-with-erasures (subsequently utilising that group's free list of erased locations) and to reuse that space than to insert a new element to the back of the colony (the default behavior when there are no previously-erased elements). If there are any erased elements in active memory blocks at the moment of insertion, colony will recycle those memory locations.

So if a block size is large, and many erasures occur but the block is not completely emptied, iterative performance might suffer due to large memory gaps between any two non-erased elements and subsequent drop in data locality and cache performance. In that scenario you may want to experiment with benchmarking and limiting the minimum/maximum sizes of the blocks, such that memory blocks are freed earlier and find the optimal size for the given use case.

7. **What is colony's Abstract Data Type (ADT)?**

   Though I am happy to be proven wrong I suspect colonies/bucket arrays are their own abstract data type. Some have suggested it's ADT is of type bag, I would somewhat dispute this as it does not have typical bag functionality such as searching based on value (you can use std::find but it's o(n)) and adding this functionality would slow down other performance characteristics. Multisets/bags are also not sortable (by means other than automatically by key value). Colony does not utilize key values, is sortable, and does not provide the sort of functionality frequently associated with a bag (e.g. counting the number of times a specific value occurs).

8. **Why must blocks be removed from the iterative sequence when empty?**

   Two reasons:

   a. Standards compliance: if blocks aren't removed then ++ and -- iterator operations become undefined in terms of time complexity, making them non-compliant with the C++ standard. At the moment they are O(1) amortized, in the reference implementation this constitutes typically one update for both skipfield and element pointers, but two if a skipfield jump takes the iterator beyond the bounds of the current block and into the next block. But if empty blocks are allowed, there could be anywhere between 1 and `std::numeric_limits<size_type>::max` empty blocks between the current element and the next. Essentially you get the same scenario as you do when iterating over a boolean skipfield. It would be possible to move these to the back of the colony as trailing blocks, or house them in a separate list or vector for future usage, but this may create performance issues if any of the blocks are not at their maximum size (see below).

   b. Performance: iterating over empty blocks is slower than them not being present, of course - but also if you have to allow for empty blocks while iterating, then you have to include a while loop in every iteration operation, which increases cache misses and code size. The strategy of removing blocks when they become empty also statistically removes (assuming randomized erasure patterns) smaller blocks from the colony before larger blocks, which has a net result of improving iteration, because with a larger block, more iterations within the block can occur before the end-of-block condition is reached and a jump to the next block (and subsequent cache miss) occurs. Lastly, pushing to the back of a colony, provided there is still space and no new block needs to be allocated, will be faster than recycling memory locations as each subsequent insertion occurs in a subsequent memory location (which is cache-friendlier) and also less computational work is necessary. If a block is removed from the iterative sequence its recyclable memory locations are also not usable, hence subsequent insertions are more likely to be pushed to the back of the colony.

9. **Why not reserve all empty memory blocks for future use during erasure, or None, rather than leaving this decision undefined by the specification?**

   The default scenario, for reasons of predictability, should be to free the memory block in most cases. However for the reasons described in the design decisions section on erase(), retaining the back block at least has performance and latency benefits. Therefore retaining no memory blocks is non-optimal in cases where the user is not using a custom allocator. Meanwhile, retaining All memory blocks is bad for performance as many small memory blocks will be retained, which decreases iterative performance due to lower cache locality. However, one perspective is that if a scenario calls for retaining memory blocks instead of deallocating them, this should be left to an allocator to manage. Otherwise you get unpredictable memory behavior across implementations, and this is one of the things that SG14 members have complained about consistently with STL implementations. This is currently an open topic for discussion.

10. **Memory block sizes - what are they based on, how do they expand, etc**

    While implementations are free to chose their own limits and strategies here, in the reference implementation

memory block sizes start from either the dynamically-defined default minimum size (8 elements, larger if the type stored is small) or an amount defined by the end user (with a minimum of 3 elements, as there is enough metadata per-block that less than 3 elements is generally a waste of memory unless the value_type is extremely large). Subsequent block sizes then increase the *total capacity* of the colony by a factor of 2 (so, 1st block 8 elements, 2nd 8 elements, 3rd 16 elements, 4th 32 elements etcetera) until the maximum block size is reached. The default maximum block size in the reference implementation is the maximum possible number that the skipfield bitdepth is capable of representing (std::numeric_limits<skipfield_type>::max()). By default the skipfield bitdepth is 16 so the maximum size of a block would be 65535 elements in that context.

The skipfield bitdepth was initially a template parameter which could be set to any unsigned integer - unsigned char, unsigned int, Uint_64, etc. Unsigned short (guaranteed to be at least 16 bit, equivalent to C++11's uint_least16_t type) was found to have the best performance in real-world testing on x86 and x86_64 platforms due to the balance between memory contiguousness, memory waste and the number of allocations. unsigned char was found to have better performance below 1000 elements and of course lower memory use. Other platforms have not been tested. Since only two values were considered useful, they've been replaced in newer versions by a `priority` parameter, which specifies whether the priority of the instantiation is memory use or performance. While this is not strictly true in the sense that unsigned char will also have better performance for under 1000 elements, it is a compromise in order to have the implementation reflect a standard which may enable other implementations which do not share the same performance characteristics.

11. **Can a colony be used with SIMD instructions?**

    No and yes. Yes if you're careful, no if you're not.
    On platforms which support scatter and gather operations via hardware (e.g. AVX512) you can use colony with SIMD as much as you want, using gather to load elements from disparate or sequential locations, directly into a SIMD register, in parallel. Then use scatter to push the post-SIMD-process values elsewhere after. On platforms which do not support this in hardware, you would need to manually implement a scalar gather-and-scatter operation which may be significantly slower.

    In situations where gather and scatter operations are too expensive, which require elements to be contiguous in memory for SIMD processing, this is more complicated. When you have a bunch of erasures in a colony, there's no guarantee that your objects will be contiguous in memory, even though they are sequential during iteration. Some of them may also be in different memory blocks to each other. In these situations if you want to use SIMD with colony, you must do the following:

    - Set your minimum and maximum group sizes to multiples of the width of your target processor's SIMD instruction size. If it supports 8 elements at once, set the group sizes to multiples of 8.
    - Either never erase from the colony, or:
        1. Shrink-to-fit after you erase (will invalidate all pointers to elements within the colony).
        2. Only erase from the back or front of the colony, and only erase elements in multiples of the width of your SIMD instruction e.g. 8 consecutive elements at once. This will ensure that the end-of-memory-block boundaries line up with the width of the SIMD instruction, provided you've set your min/max block sizes as above.

    Generally if you want to use SIMD without gather/scatter, it's probably preferable to use a vector or an array.

Appendix D - Specific responses to previous committee feedback

1. **"Why not 'bag'? Colony is too selective a name."**

    'bag' is problematic partially because it has been synonymous with a multiset (and colony is not one of those) in both computer science and mathematics since the 1970s, and partially because it's a bit vague - it doesn't describe how the container works. However I accept that it is a familiar name and describes a similar territory, for most programmers and will accept that as a id if needed. 'colony' is an intuitive name if you understand the container, and allows for easy conveyance of how it functions internally (colony = human colony/ant colony etc, memory blocks = houses, elements = people/ants in the houses who come and go). The claim that the use of the word is selective in terms of its meaning, is also true for vector, set, 'bag', and many other C++ names.

2. **"Unordered and no associative lookup, so this only supports use cases where you're going to do something to every element."**

As noted the container was originally designed for highly object-oriented situations where you have many elements in different containers linking to many other elements in other containers. This linking can be done with pointers or iterators in colony (insert returns an iterator which can be dereferenced to get a pointer, pointers can be converted into iterators with the supplied functions (for erase etc)) and because pointers/iterators stay stable regardless of insertion/erasure, this usage is unproblematic. You could say the pointer is equivalent to a key in this case (but without the overhead). That is the first access pattern, the second is straight iteration over the container, as you say. Secondly, the container does have (typically better than O(n)) advance/next/prev implementations, so multiple elements can be skipped.

3. **"Do we really need the Priority template parameter?"**

   While technically a non-binding request, this parameter promotes the use of the container in heavily memory-constrained environments like embedded programming. In the context of the reference implementation this means switching the skipfield type from unsigned short to unsigned char, in other implementations it could mean something else. See more explanation in V. Technical Specifications. Unfortunately this parameter also means `operator=`, swap and some other functions won't work between colonies of the same type but with differing priorities.

4. **"Prove this is not an allocator"**

   I'm not really sure how to answer this, as I don't see the resemblance, unless you count maps, vectors etc as being allocators also. The only aspect of it which resembles what an allocator might do, is the memory re-use mechanism. It would be impossible for an allocator to perform a similar function while still allowing the container to iterate over the data linearly in memory, preserving locality, in the manner described in this document.

5. **"If this is for games, won't game devs just write their own versions for specific types in order to get a 1% speed increase anyway?"**

   This is true for many/most AAA game companies who are on the bleeding edge, but they also do this for vector etc, so they aren't the target audience of std:: for the most part; sub-AAA game companies are more likely to use third party/pre-existing tools. As mentioned earlier, this structure (bucket-array-like) crops up in many, many fields, not just game dev. So the target audience is probably everyone other than AAA gaming, but even then, it facilitates communication across fields and companies as to this type of container, giving it a standardized name and understanding.

6. **"Is there active research in this problem space? Is it likely to change in future?"**

   The only current analysis has been around the question of whether it's possible for this specification to fail to allow for a better implementation in future. This is unlikely given the container's requirements and how this impacts on implementation. Bucket arrays have been around since the 1990s, there's been no significant innovation in them until now. I've been researching/working on colony since early 2015, and while I can't say for sure that a better implementation might not be possible, I am confident that no change should be necessary to the specification to allow for future implementations, if it is done correctly.

   The requirement of allowing no reallocations upon insertion or erasure, truncates possible implementation strategies significantly. Memory blocks have to be independently allocated so that they can be removed (when empty) without triggering reallocation of subsequent elements. There's limited numbers of ways to do that and keep track of the memory blocks at the same time. Erased element locations must be recorded (for future re-use by insertion) in a way that doesn't create allocations upon erasure, and there's limited numbers of ways to do this also. Multiple consecutive erased elements have to be skipped in O(1) time, and again there's limits to how many ways you can do that. That covers the three core aspects upon which this specification is based. See IV. Design Decisions for the various ways these aspects can be designed.

   The time complexity of updates to whatever erased-element skipping mechanism is used should, I think, be left implementation-defined, as defining time complexity may obviate better solutions which are faster but are not necessarily O(1). These updates would likely occur during erasure, insertion, splicing and container copying.

7. **Why not iterate across the memory blocks backwards to find the first block with erasures to reuse, during insert?**

While this would statistically ensure that smaller blocks get deallocated first due to becoming empty faster than later blocks, it introduces uncertain latency issues during insert, particularly when custom memory block sizes are used and the number of elements is large. With the current implementation there is an intrusive list of blocks with erasures, and within each block's metadata there's a free list of skipblocks. When reusing the current head of the intrusive list determines the block, and the current head of that block's free list determines the skipblock to be reused. This means that the most recently erased element will be the first to reused. This works out well for two reasons: currently-contiguous sequences of elements will tend to stay that way, helping cache coherence, and when elements are erased and inserted in sequence those erased memory locations will tend to be already in the cache when inserting. Lastly, this structure involves a minimum of branching and checks, resulting in minimal latency during insertion and erasure.

Appendix E - Typical game engine requirements

Here are some more specific requirements with regards to game engines, verified by game developers within SG14:

a. Elements within data collections refer to elements within other data collections (through a variety of methods - indices, pointers, etc). These references must stay valid throughout the course of the game/level. Any container which causes pointer or index invalidation creates difficulties or necessitates workarounds.

b. Order is unimportant for the most part. The majority of data is simply iterated over, transformed, referred to and utilized with no regard to order.

c. Erasing or otherwise "deactivating" objects occurs frequently in performance-critical code. For this reason methods of erasure which create strong performance penalties are avoided.

d. Inserting new objects in performance-critical code (during gameplay) is also common - for example, a tree drops leaves, or a player spawns in an online multiplayer game.

e. It is not always clear in advance how many elements there will be in a container at the beginning of development, or at the beginning of a level during play. Genericized game engines in particular have to adapt to considerably different user requirements and scopes. For this reason extensible containers which can expand and contract in realtime are necessary.

f. Due to the effects of cache on performance, memory storage which is more-or-less contiguous is preferred.

g. Memory waste is avoided.

std::vector in its default state does not meet these requirements due to:

1. Poor (non-fill) single insertion performance (regardless of insertion position) due to the need for reallocation upon reaching capacity
2. Insert invalidates pointers/iterators to all elements
3. Erase invalidates pointers/iterators/indexes to all elements after the erased element

Game developers therefore either develop custom solutions for each scenario or implement workarounds for vector. The most common workarounds are most likely the following or derivatives:

1. Using a boolean flag or similar to indicate the inactivity of an object (as opposed to actually erasing from the vector). Elements flagged as inactive are skipped during iteration.

   Advantages: Fast "deactivation". Easy to manage in multi-access environments.
   Disadvantages: Can be slower to iterate due to branching.

2. Using a vector of data and a secondary vector of indexes. When erasing, the erasure occurs only in the vector of indexes, not the vector of data. When iterating it iterates over the vector of indexes and accesses the data from the vector of data via the remaining indexes.

   Advantages: Fast iteration.
   Disadvantages: Erasure still incurs some reallocation cost which can increase jitter.

3. Combining a swap-with-back-element-and-pop approach to erasure with some form of dereferenced lookup system to enable contiguous element iteration (sometimes called a 'packed array': http://bitsquid.blogspot.ca/2011/09/managing-decoupling-part-4-id-lookup.html).
   Advantages: Iteration is at standard vector speed.
   Disadvantages: Erasure will be slow if objects are large and/or non-trivially copyable, thereby making swap costs large. All link-based access to elements incur additional costs due to the dereferencing system.

Colony brings a more generic solution to these contexts. While some developers, particularly AAA developers, will almost always develop a custom solution for specific use-cases within their engine, I believe most sub-AAA and indie developers are more likely to rely on third party solutions. Regardless, standardising the container will allow for

greater cross-discipline communication.

Appendix F - Time complexity requirement explanations

Insert (single): O(1)

One of the requirements of colony is that pointers to non-erased elements stay valid regardless of insertion/erasure within the container. For this reason the container must use multiple memory blocks. If a single memory block were used, like in a std::vector, reallocation of elements would occur when the container expanded (and the elements were copied to a larger memory block). Instead, colony will insert into existing memory blocks when able, and create a new memory block when all existing memory blocks are full. This keeps insertion at O(1).

Insert (multiple): O(N)

Multiple insertions may allow an implementation to reserve suitably-sized memory blocks in advance, reducing the number of allocations necessary (whereas singular insertion would generally follow the implementation's block growth pattern, possibly allocating more than necessary). However when it comes to time complexity it has no advantages over singular insertion, is linear to the number elements inserted.

Erase (single): O(1)

Erasure is a simple matter of destructing the element in question and updating whatever data is associated with the erased-element skipping mechanism eg. the skipfield. Since we use a skipping mechanism to avoid erasures during iterator, no reallocation of subsequent elements is necessary and the process is O(1). Additionally, when using a Low-complexity jump-counting pattern the skipfield update is also always O(1).

Note: When a memory block becomes empty of non-erased elements it must be freed to the OS (or stored for future insertions, depending on implementation) and removed from the colony's sequence of memory blocks. It it was not, we would end up with non-O(1) iteration, since there would be no way to predict how many empty memory blocks there would be between the current memory block being iterated over, and the next memory block with non-erased (active) elements in it.

Erase (multiple): O(N) for non-trivially-destructible types, for trivially-destructible types between O(1) and O(N) depending on range start/end, approximating O(log n) average

In this case, where the element is non-trivially destructible, the time complexity is O(N), with infrequent deallocation necessary from the removal of an empty memory block as noted above. However where the elements are trivially-destructible, if the range spans an entire memory block at any point, that block and it's metadata can simply be removed without doing any individual writes to it's metadata or individual destruction of elements, potentially making this a O(1) operation.

In addition (when dealing with trivially-destructible types) for those memory blocks where only a portion of elements are erased by the range, if no prior erasures have occurred in that memory block you may be able to erase that range in O(1) time, as, for example, if you are using a skipfield there will be no need to check the skipfield within the range for previously erased elements. The reason you would need to check for previously erased elements within that portion's range is so you can update the metadata for that memory block to accurately reflect how many non-erased elements remain within the block. The non-erased element-count metadata is necessary because there is no other way to ascertain when a memory block is empty of non-erased elements, and hence needs to be removed from the colony's iteration sequence. The reasoning for why empty memory blocks must be removed is included in the Erase(single) section, above.

However in most cases the erase range will not perfectly match the size of all memory blocks, and with typical usage of a colony there is usually some prior erasures in most memory blocks. So, for example, when dealing with a colony of a trivially-destructible type, you might end up with a tail portion of the first memory block in the erasure range being erased in O(N) time, the second and intermediary memory block being completely erased and freed in O(1) time, and only a small front portion of the third and final memory block in the range being erased in O(N) time. Hence the time complexity for trivially-destructible elements approximates O(log n) on average, being between O(1) and O(N) depending on the start and end of the erasure range.

std::find: O(N)

This relies on basic iteration so is O(N).

splice: O(1)

Colony only does full-container splicing, not partial-container splicing (use range-insert with std::make_move_iterator to achieve the latter, albiet with the loss of pointer validity to the moved range). When splicing, the memory blocks from the source colony are transferred to the destination colony without processing the individual elements. These blocks may either be placed at the front of the colony or the end, depending on how full the source back block is compared to the destination back block. If the destination back block is more full ie. there is less unused space in it, it is better to put it at the beginning of the source block - as otherwise this creates a larger gap to skip during iteration which in turn affects cache locality. If there are unused element memory spaces at the back of the destination container (ie. the final memory block is not full) and a skipfield is used, the skipfield nodes corresponding to those empty spaces must be altered to indicate that these are skipped elements.

Iterator operators ++ and --: O(1) amortized

Generally the time complexity is O(1), and if a skipfield pattern is used it must allow for O(1) skipping of multiple erased elements. However every so often iteration will involve a transistion to the next/previous memory block in the colony's sequence of blocks, depending on whether we are doing ++ or --. At this point a read of the next/previous memory block's corresponding skipfield would be necessary, in case the front/back element(s) in that memory block are erased and hence skipped. So for every block transition, 2 reads of the skipfield are necessary instead of 1. Hence the time complexity is O(1) amortized.

If skipfields are used they must be per-element-memory-block and independent of subsequent/previous memory blocks, as otherwise you would end up with a vector for a skipfield, which would need a range erased every time a memory block was removed from the colony (see notes under Erase above), and reallocation to a larger skipfield memory block when a colony expanded. Both of these procedures carry reallocation costs, meaning you could have thousands of skipfield nodes needing to be reallocated based on a single erasure (from within a memory block which only had one non-erased element left and hence would need to be removed from the colony). This is unacceptable latency for any field involving high timing sensitivity (all of SG14).

begin()/end(): O(1)

For any implementation these should generally be stored as member variables and so returning them is O(1).

advance/next/prev: between O(1) and O(n), depending on current iterator location, distance and implementation. Average for reference implementation approximates O(log N).

The reasoning for this is similar to that of Erase(multiple), above. Complexity is dependent on state of colony, position of iterator and length of `distance`, but in many cases will be less than linear. It is necessary in a colony to store metadata both about the capacity of each block (for the purpose of iteration) and how many non-erased elements are present within the block (for the purpose of removing blocks from the iterative chain once they become empty). For this reason, intermediary blocks between the iterator's initial block and its final destination block (if these are not the same block, and if the initial block and final block are not immediately adjacent) can be skipped rather than iterated linearly across, by subtracting the "number of non-erased elements" metadata from `distance` for those blocks.

This means that the only linear time operations are any iterations within the initial block and the final block. However if either the initial or final block have no erased elements (as determined by comparing whether the block's capacity metadata and the block's "number of non-erased elements" metadata are equal), linear iteration can be skipped for that block and pointer/index math used instead to determine distances, reducing complexity to constant time. Hence the best case for this operation is constant time, the worst is linear to the distance.

distance: between O(1) and O(n), depending on current iterator location, distance and implementation. Average for reference implementation approximates O(log N).

The same considerations which apply to advance, prev and next also apply to distance - intermediary blocks between iterator1 and iterator2's blocks can be skipped in constant time, if they exist. iterator1's block and iterator2's block (if these are not the same block) must be linearly iterated across using ++ unless either block has no erased elements, in which case the operation becomes pointer/index math and is reduced to constant time for that block. In addition, if iterator1's block is not the same as iterator2's block, and iterator2 is equal to end() or (end() - 1), or is the last element in that block, iterator2's block's elements can also counted from the metadata rather than iteration.

Appendix G - Why not constexpr?

I am somewhat awkwardly forced into a position where I have to question and push back against the currently-unsubstantiated enthusiasm around constexpr containers and functions. At the time of writing there are no compilers

which both support constexpr non-trivial destructors and also have a working implementation of a constexpr container. And until that is remedied, we won't really know what we're dealing with. My own testing in terms of making colony constexpr has not been encouraging. 2% performance decrease in un-altered benchmark code is common, and I suspect the common cause of this is caching values from compile-time when it is cheaper to calculate them on-the-fly than to return them from main memory. This suspicion is based on the substantial increases in executable size in the constexpr versions.

For an example of the latter, think about size() in std::vector. This can be calculated in most implementations by (vector.end_iterator.pointer - vector.memory_block), both of which will most likely be in cache at the time of calling size(). That's if size isn't a member variable. Calculating a minus operation on stuff that's already in cache is about 100x faster than making a call out to main memory for a compile-time-stored value of this function, if that is necessary.

None of which is an issue if a container is being entirely used within a constexpr function which has been determined to be evaluated at compile time. The problems occur when constexpr containers are used in runtime code, but certain functions such as size() are determined to be able to be evaluated at compile time, and therefore have their results cached. This is not an okay situation, performance-wise. If there were a mechanism which specified that for a given class instance, it's constexpr functions may **not** be evaluated at compile time, then I would give the go-ahead. Similarly if there were a rule which stated that a class instance's member functions may only be evaluated at compile time if the class instance is instantiated and destructed at compile time, I would give the go-ahead. This is not our situation.

Constexpr function calls:

a. shift the responsibility of storage of pre-calculated results from the programmer to the compiler, and remove the ability for the programmer to think about the cache locality/optimal storage of precalculated results

b. shift the decision of whether or not to evaluate at runtime/compile-time to the compiler, where in some cases doing it at compile-time and storing the result may decrease performance (see the Doom 3 BFG edition technical notes for their experience with pre-calc'ing meshes vs calculating them on the fly)

c. may dramatically increase code size/file size in some cases if the return results of constexpr functions are large

d. may dramatically increase compile times

e. create the potential for cache pollution when constexpr functions return large amounts of data, or when functions returning small amounts of data are called many times

Given this, and the performance issues mentioned above, I am reluctant to make colony constexpr-by-default. Time may sort these issues out, but I am personally happier for std::array and std::vector to be the "canaries in the coalmine" here. Certainly I won't be giving the go-ahead on any change that produces, or can produce, on current compilers, a 2% performance decrease in runtime code. Though I acknowledge the functionality of constexpr code may be useful to many.